Automatic extraction of commonsense information from the web using contextual clues

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Abstract: There have been numerous attempts to create Artificial Intelligence agents that can reason about the world the same way humans do. It has become clear that in order for that to happen the agent must have the common knowledge, or "common sense", about the world we live in. With humans, this information is acquired naturally during our development stages, through both knowledge acquisition and reasoning based on what is already known. Computer agents, however, have no inherent mechanisms to acquire "common sense" knowledge or to derive inferences based on it. These mechanisms have to be supplied by the creators of the agent. Many projects are currently involved in manually providing such taxonomies but this process is costly and slow. The World Wide Web, however, provides a ready source of common sense information that we can use. In this paper we focus on the possibility of automatically identifying "common sense" knowledge from unstructured text sources. Much of the "common sense" is stored as generic statements of the kind "Dogs are canines". Thus we identify such statements by the extent of their "genericity" and contextual clues extracted from the text. Once we have perfected our approach, we plan to use the collected common sense to semantically annotate the World Wide Web.